

22 HEATING EFFECT OF ELECTRIC CURRENT

POWER
 $P = \frac{dH}{dt} = VI = \frac{V^2}{R} = I^2R$

ELECTRIC KETTLE
 Time taken for first coil - t_1 , time taken for second coil - t_2
 If they are connected in series
 $t_s = t_1 + t_2$
 If they are connected in parallel
 $t_p = \frac{t_1 t_2}{t_1 + t_2}$

BULB
 $P_{rated} = \frac{V_{rated}^2}{R} \Rightarrow R = \frac{V_{rated}^2}{P_{rated}}$

CONNECTED IN SERIES

 $P_{combined} = IR^2$
 $P_{combined} \propto R$
 Brightness $\propto R$

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$P_{dissipated} \propto \frac{V_{rated}^2}{P_{rated}}$
 $P_{et} \text{ brightness} \propto \frac{1}{P_{rated}}$

CONNECTED IN PARALLEL

 $P_{total} = \frac{V_{total}^2}{R}$
 $P_s = \frac{V_s^2}{R_s} + \frac{V_s^2}{R_s}$
 $\Rightarrow P_{brightness} \propto \frac{1}{(V_{total})^2} \Rightarrow P_{brightness} \propto \frac{1}{(V_{total})^2}$
 $P_{et} \text{ (brightness)} \propto P_{rated}$
 if $(P_1)_e > (P_2)_e$
 $\Rightarrow \text{brightness } (P_1)_e > (P_2)_e$

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$P_{R2} > P_{R1}$

COMBINATION OF BULBS

SERIES

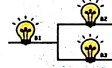
 P_1 P_2
TOTAL POWER
 $P = \frac{P_1 P_2}{P_1 + P_2}$

PARALLEL

 $P = P_1 + P_2$

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FUSED BULB


 If bulb 2 is fused then,
 for bulb 1
 R I
 V I
 P I
 B I

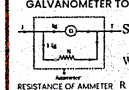
if bulb is added in parallel -
 for example if bulb 2 is added in parallel to 1 then

 for bulb 2
 R I
 V I
 P I
 B I

26 CONVERSION OF GALVANOMETER

Current sensitivity
 $S_i = \frac{\theta}{I}$
 where, θ = angle of deflection in galvanometer
 I = corresponding current in galvanometer
 Unit: $\frac{\text{degree}}{\text{ampere}}$ or $\frac{\text{rad}}{\text{A}}$

Voltage sensitivity
 $S_v = \frac{\theta}{V}$
 where, θ = angle of deflection in galvanometer
 V = corresponding voltage across galvanometer
 Unit: $\frac{\text{degree}}{\text{volt}}$ or $\frac{\text{rad}}{\text{V}}$

GALVANOMETER TO AMMETER

 Where $n = \frac{I}{I_g}$
 RESISTANCE OF AMMETER $R = \frac{G}{G + S}$
 RESISTANCE OF IDEAL AMMETER = 0

GALVANOMETER TO VOLTMETER

 $V = IG + I_g R$
 $V = \frac{I_g}{n} + I_g R$
 RESISTANCE OF VOLTMETER = 0 + R
 RESISTANCE OF IDEAL VOLTMETER = ∞

Biot-Savart's Law

$$dB \propto \frac{Idl \sin \theta}{r^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$

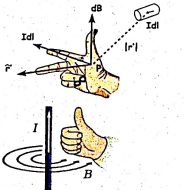
μ_0 → magnetic permeability of free space or vacuum
 $\mu_0 = 4\pi \times 10^{-7}$

In vector form,

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \vec{r}}{r^3}$$

$d\vec{B}$ is perpendicular to both $d\vec{l}$ and $\vec{r}$. By using right hand screw rule we can find direction of magnetic field

Here, B is into the plane



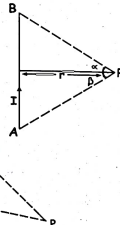
Magnetic field circulates around the current carrying wire

Formula of Field due to straight wire

1) At point P

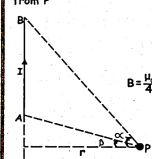
B is into the plane

$$B = \frac{\mu_0 I}{4\pi r} (\sin \alpha + \sin \beta), \otimes$$



2)

Extend AB downwards and draw a perpendicular from P



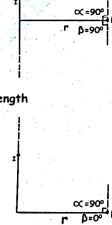
$$B = \frac{\mu_0 I}{4\pi r} (\sin \alpha - \sin \beta), \otimes$$

3) Wire of infinite length

$\alpha = 90^\circ$ and $\beta = 90^\circ$

$$B = \frac{\mu_0 I}{4\pi r} (\sin 90^\circ + \sin 90^\circ)$$

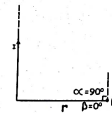
$$= \frac{\mu_0 I}{2\pi r}, \otimes$$



4) Wire of semi infinite length

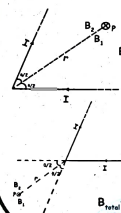
$$B = \frac{\mu_0 I}{4\pi r} (\sin 90^\circ + \sin 0^\circ)$$

$$= \frac{\mu_0 I}{4\pi r}$$



$$B(P_1) = B(P_2) = 0$$

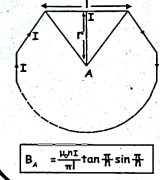
Field Due to bent wire



$$B_{\text{total}} = B_1 + B_2 = 2 \times \frac{\mu_0 I}{4\pi r} \frac{1 + \cos \theta}{2}$$

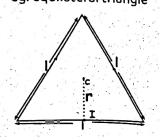
$$B_{\text{total}} = B_1 + B_2 = 2 \times \frac{\mu_0 I}{4\pi r} \frac{1 - \cos \theta}{2}$$

Field due to polygon



$$B_A = \frac{\mu_0 I}{\pi l} \tan \frac{\pi}{n} \sin \frac{\pi}{n}$$

eg: equilateral triangle

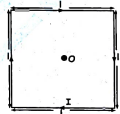


$$B_A = \frac{9}{2} \frac{\mu_0 I}{\pi l}$$

If a total length l is bent as an equilateral triangle, each side has length $l/3$ then,

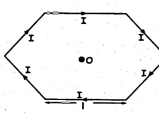
$$B' = \frac{27}{2} \frac{\mu_0 I}{\pi l}$$

Square



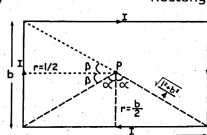
$$B = \frac{2\sqrt{2} \mu_0 I}{\pi l}$$

Hexagon



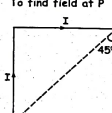
$$B = \frac{3\sqrt{3} \mu_0 I}{\pi l}$$

Rectangle



$$B_P = \frac{2\mu_0 I}{\pi} \left(\frac{\text{diagonal}}{\text{area}} \right) = \frac{2\mu_0 I}{\pi} \frac{r}{ab}$$

Note:- To find field at P



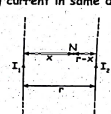
Here P is a point at vertex

$$B_P = \frac{\mu_0 I}{2\pi l}$$

Neutral Points

Points at which magnetic field becomes zero are called neutral points.
 Case 1: Parallel wires carrying current in same direction (I_1, I_2)

$$x = \frac{I_1 r}{I_1 + I_2}$$

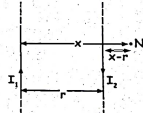


If y is the distance from 2nd conductor then

$$y = \frac{I_1 r}{I_1 + I_2}$$

Case 2: Parallel wires carrying currents in opposite direction (I_1, I_2)

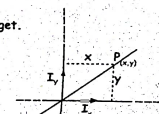
$$x = \frac{I_1 r}{I_1 - I_2}$$



Case 3: Wires perpendicular to each other
 Field will be zero on all points of the line OP

Comparing with $y = mx + c$ we get,

$$\text{slope } m = \frac{I_1}{I_2}$$

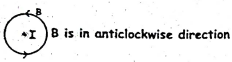


MOVING CHARGES AND MAGNETISM

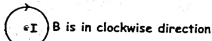
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Wire perpendicular to plane

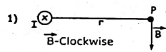
- 1) Current directed out of plane



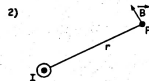
- 2) Current directed into the plane



Direction of field

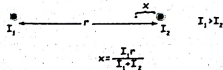


$$B = \frac{\mu_0 I}{2\pi r}$$

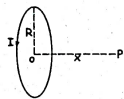


$$B = \frac{\mu_0 I}{2\pi r}$$

Neutral Point



Field due to a circular ring & Field at the axis of a ring



$$B_x = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

Every current carrying loop acts as a magnetic dipole

$$\vec{M} = I\vec{A}$$

Direction of A is determined using right hand thumb rule

$$\text{Here, } \vec{M} = I\vec{A} \Rightarrow M = I\pi R^2$$

$$B = \frac{\mu_0}{4\pi} \frac{2M}{(x^2 + R^2)^{3/2}}$$

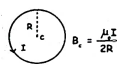
If current is flowing in anticlockwise direction



If current is flowing in clockwise direction



Field at the centre of ring

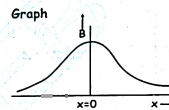


$$B_c = \frac{\mu_0 I}{2R}$$

Field at the axis in terms of field at center

$$B_{axis} = \frac{B_c}{\left(1 + \frac{x^2}{R^2}\right)^{3/2}}$$

Graph

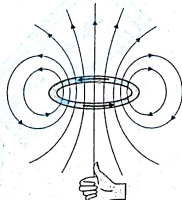


Percentage change in field with respect to centre for the points on the axis

$$\% \text{ change in } B = 1 - \left(1 + \frac{x^2}{R^2}\right)^{-3/2} \times 100\%$$

$$\text{For change } < 10\% = \left(\frac{3}{2} \times \frac{x^2}{R^2}\right) \times 100\%$$

Magnetic field lines for a current loop



If there are N loops,

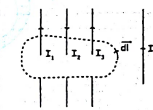
$$B = \frac{\mu_0 N I R^2}{2(x^2 + R^2)^{3/2}}$$

Ampere's Circuital Law

The line integral of magnetic field over a closed loop is equal to μ_0 times the total current enclosed by the loop

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Example



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 [I_1 - I_2 + I_3]$$

Solid Cylinder



Outside : $r > R$

$$B_{out} = \frac{\mu_0 I}{2\pi r}$$

$$B_{out} \propto \frac{1}{r}$$

On the surface $r = R$

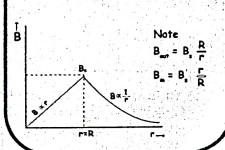
$$B_s = \frac{\mu_0 I}{2\pi R}$$

Inside ($r < R$)

$$B_{in} = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$$

$$B_{in} \propto r$$

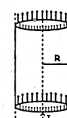
Graph



$$\text{Note } B_{out} = B_s \frac{R}{r}$$

$$B_{in} = B_s \frac{r}{R}$$

Hollow cylinder (pipe)



Outside : $r > R$

$$B_{out} = \frac{\mu_0 I}{2\pi r}$$

$$B_{out} \propto \frac{1}{r}$$

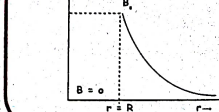
On the surface : $r = R$

$$B_s = \frac{\mu_0 I}{2\pi R}$$

Inside : $r < R$

$$B_{in} = 0$$

Graph



MOVING CHARGES AND MAGNETISM

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MOVING CHARGES AND MAGNETISM

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Force on current carrying Conductor in magnetic field

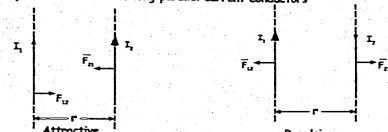
consider a small element $d\vec{l}$ of the conductor
 $d\vec{F} = I(d\vec{l} \times \vec{B})$
 To find resulting force
 $\int d\vec{F} = I \int (d\vec{l} \times \vec{B})$
 $= I \int d\vec{l} \times \vec{B}$
 In uniform field
 $\vec{F} = I(\int d\vec{l}) \times \vec{B} = I\vec{l} \times \vec{B}$

The above equation can be used in the following situations

- 1)  $\vec{F} = I(\vec{l} \times \vec{B})$
 $= I l B \sin \theta$
- 2) $\vec{l}$ perpendicular to $\vec{B}$
 $\theta = 90^\circ$
 $F_{max} = I l B$
 For closed loop in uniform field
 Here, $\vec{l}_{net} = 0$
 $\therefore \vec{F} = 0$

Force between parallel conductors

1) Force between two long parallel current conductors



Parallel currents attract each other. Anti parallel currents repel each other.
 Force per unit length of conductor
 $F_{12}/F_{21} = \frac{\mu_0}{4\pi} \frac{I_1 I_2}{r}$
 Net force on 'l' length of conductor $= \frac{\mu_0}{4\pi} \frac{2 I_1 I_2}{r} \times l$

Motion of charged particle in Crossed Electric and Magnetic field $\vec{E} \perp \vec{B}$ (velocity selector)

$$\vec{F} = q(\vec{E} + (\vec{V} \times \vec{B})) = \vec{F}_E + \vec{F}_B$$

$$\begin{aligned} \vec{F}_E &= q\vec{E} = qE\hat{j} \\ \vec{F}_B &= q(\vec{V} \times \vec{B}) = q(V\hat{i} \times B\hat{k}) \\ &= -qV B \hat{j} \\ \therefore \vec{F} &= q(E - VB)\hat{j} \end{aligned}$$

If $v = E/B$, $F_{net} = 0$
 Charge moves in a Straight line

Thus, electric and magnetic forces are in opposite directions as shown in figure.

Lorentz force

A charge q in an electric field $\vec{E}$ experiences the electric force

$$\vec{F} = q\vec{E}$$

The magnetic force experienced by the charge q moving with velocity $\vec{V}$ in the magnetic field $\vec{B}$ is given by

$$\vec{F}_m = q(\vec{V} \times \vec{B})$$

The total force, or the Lorentz force, experienced by the charge q due to both electric and magnetic field is given by

$$\vec{F} = q[\vec{E} + (\vec{V} \times \vec{B})]$$

Magnetic dipole moment of a revolving electron

$$\begin{aligned} I &= \frac{q}{T} \\ T &= \frac{2\pi r}{v} \\ I &= \frac{qv}{2\pi r} \end{aligned}$$



There will be a magnetic moment, usually denoted by μ associated with this circulating current

$$\mu = I \pi r^2 = \frac{eVr}{2}$$

$$\mu = \frac{e}{2m_e} (m_e v r) = \frac{e}{2m_e} L \quad \vec{L} = \vec{r} \times \vec{p} = mvr \sin \theta = mv r$$

In vector form

$$\vec{\mu} = \frac{-e}{2m_e} \vec{L}$$

The negative sign indicates that the angular momentum of the electron is opposite in direction to the magnetic moment

The ratio of magnetic moment to the angular momentum is called gyromagnetic ratio

$$\frac{\mu}{L} = \frac{e}{2m_e}$$

Its value is a constant and is equal to $8.8 \times 10^{-10} \text{ C/kg}$ for an electron

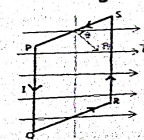
According to Bohr's quantization condition, angular momentum assumes a discrete set of values, namely,

$$L = \frac{nh}{2\pi} \text{ where } n = 1, 2, 3, \dots \quad h = \text{Planck's constant}$$

Taking $n=1$, we get,

$$\mu_{min} = \frac{e}{4\pi m_e} h = 9.27 \times 10^{-24} \text{ Am}^2$$

Torque on a current loop in a uniform magnetic field



Then $\tau = NIAB \sin \theta$

Special cases:

i) When $\theta = 0^\circ$

$\tau = 0$, i.e., the torque is minimum when the plane of the loop is perpendicular to the magnetic field

ii) When $\theta = 90^\circ$

$\tau = NIAB$, i.e., the torque is maximum when the plane of the loop is parallel to the magnetic field. Thus,

$$\tau_{max} = NIAB$$

Moving Coil Galvanometer (MCG)

$$\tau = NIAB \sin \theta$$

Sensitivity of a Galvanometer

$$\text{Current sensitivity, } I_s = \frac{\Phi}{I} = \frac{NAB}{K}$$

Where K — torsional constant

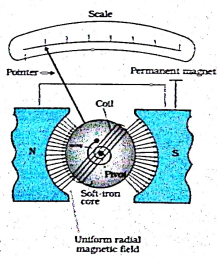
Φ — deflection of galvanometer

Voltage Sensitivity

$$\text{Voltage Sensitivity, } V_s = \frac{\Phi}{V} = \frac{\Phi}{IR}$$

$$= \frac{NAB}{KR}$$

$$\text{Voltage Sensitivity} = \frac{\text{Current Sensitivity}}{R}$$



Note:

1) The constant, $\frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.

2) Figure of merit of a galvanometer

$$G = \frac{I}{\Phi} = \frac{K}{NAB}$$

THE MAGNETIC FIELD LINES

1. The magnetic field lines of a magnet form continuous closed loops
2. The tangent to the field lines at a given point represents the direction of the net magnetic field B at that point
3. The larger no. of field lines $\rightarrow$ stronger B
4. Do not intersect

MAGNETIC DIPOLE MOMENT (M)

$M = m \cdot l$
 m $\rightarrow$ pole strength
 l $\rightarrow$ straight line distance b/w poles
 Direction from south pole to N pole
 Unit of $M \rightarrow Am^2$
 Unit of $m \rightarrow Am$

BAR MAGNET TO DIFFERENT SHAPES

$M_{\text{new}} = M \sin(\frac{\theta}{2})$
 $l = \frac{l \sin(\frac{\theta}{2})}{\sin(\frac{\theta}{2})}$

RESULTANT DIPOLE MOMENT

$M_{\text{net}} = \sqrt{M_1^2 + M_2^2 + 2M_1M_2 \cos \theta}$

CUTTING OF BAR MAGNET

LENGTHWISE / TRANSVERSE
 $\frac{l}{2}$ $\frac{l}{2}$
 Pole strength $\rightarrow$ same
 length $\rightarrow$ reduce to half
 $M_{\text{new}} = \frac{ml}{2}$

HORIZONTAL
 $\frac{l}{2}$ $\frac{l}{2}$
 Pole strength $\rightarrow$ reduce to half
 length $\rightarrow$ same
 $M_{\text{new}} = \frac{ml}{2}$

FACTS

- A) Magnetic monopoles does not exist
- B) A solenoid and bar magnet produce similar magnetic fields

THE ELECTROSTATIC ANALOG (Help from electrostatics to magnetism)

SHORT ELECTRIC DIPOLE
 q_1 q_2
 $P = q \cdot d$
COULOMB'S LAW
 $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$
AXIAL & EQUATORIAL LINE OF DIPOLE
 $E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2P}{r^3}$
 $E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3}$
 $E_{\text{axial}} = -2E_{\text{eq}}$

SHORT MAGNETIC DIPOLE
 m_1 m_2
 $M = m \cdot l$
 $F_m = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^2}$
 $B_{\text{axial}} = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$
 $B_{\text{eq}} = \frac{\mu_0}{4\pi} \frac{M}{r^3}$
 $B_{\text{axial}} = -2B_{\text{eq}}$

POTENTIAL AT ANY GENERAL POINT

$V = \frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2}$
 $V_m = \frac{\mu_0}{4\pi} \frac{M \cos \theta}{r^2}$

TORQUE

1) $F_{\text{net}} = 0$
 $\tau = P \times E = pE \sin \theta$
 $\tau = M \times B = MB \sin \theta$

WORK DONE IN ROTATING A DIPOLE

1. $W = PE (\cos \theta_1 - \cos \theta_2)$
 Maximum work done is from $\theta_1 = 0^\circ$ to $\theta_2 = 180^\circ$
 $W_s = MB (\cos \theta_1 - \cos \theta_2)$

POTENTIAL ENERGY

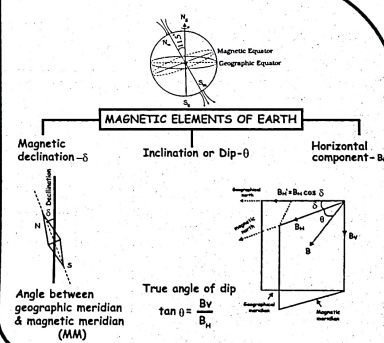
$U = -P \cdot E$
 $U_m = -M \cdot B$
 $\theta_1 = 0^\circ$ Stable position; $\theta = 180^\circ$ Unstable position

MAGNETISM AND GAUSS' LAW

"The net magnetic flux through any closed surface is zero"
 $\oint \vec{B} \cdot d\vec{s} = 0$

NOTE: "The simplest magnetic element is a magnetic dipole or a current loop". Magnetic monopoles do not exist.

THE EARTH'S MAGNETISM



MAGNETISM AND MATTER

APPARENT ANGLE OF DIP

Inclination of magnetic needle in plane other than magnetic meridian

$\tan \delta' = \frac{\tan \theta}{\cos \phi}$
 δ' Apparent angle of dip
 θ true angle of dip
 ϕ Angle between MM and the plane other than MM

RELATION BETWEEN TWO FALSE ANGLE OF DIPS (δ_1, δ_2) IN MUTUALLY PERPENDICULAR PLANES AND TRUE ANGLE OF DIP (θ)

$\cot^2 \delta_1 + \cot^2 \delta_2 = \cot^2 \theta$

FACTS

1. Declination is greater at poles and smaller near equator
2. Angle of dip is maximum at poles and minimum at equator

COMPASS NEEDLE AND DIP NEEDLE

1. A compass needle at the North pole can point along any direction.
2. A dip needle at the north pole points down and at South pole points straight up.

TIME PERIOD

of a magnetic dipole in uniform magnetic field
 $T = 2\pi \sqrt{\frac{I}{MB}}$
 I - Moment of Inertia of the body
 M - Magnetic dipole moment
 B - Magnetic field

Frequency $\nu = \frac{1}{2\pi} \sqrt{\frac{MB}{I}}$ To find B , $B = \frac{4\pi^2 I}{MT^2}$

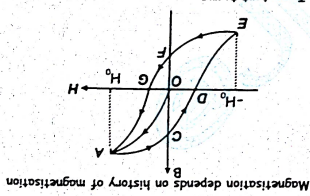
MAGNETISM

ELECTROMAGNETS
Materials should have high permeability
Soft iron is used
Used in electric bells, Loudspeakers, telephone diaphragms, heavy cranes to lift machinery

Permanent magnets
should have
1. High retentivity
2. High permeability
Steel is used for making permanent magnets
Steel
Higher coercivity
Smaller retentivity than steel
Higher coercivity than steel
soft iron

Area of hysteresis loop
Smaller for soft iron
Higher for steel
B-H curve signifies the energy loss/heat loss in the process and is proportional to the area of the loop.

Important terms
Retentivity - OC - Residual magnetism
Coercivity - OD - Demagnetising process
1. High coercivity - Hard substance - Steel
2. Low coercivity - Soft substance - Soft iron



f. At high temperature, a ferromagnetic substance becomes paramagnetic
Curie's temperature
$$\chi = \frac{1}{T - T_c} \quad (T > T_c)$$

e. Important
 $\chi \gg 1$
 $\mu_r \gg 1$
3. Ferromagnetic substances
a. Strongly attracted by a magnet
b. Eg : Fe, Co, Ni, Cd, Fe₃O₄
c. Individual atoms possess permanent magnetic moment and magnetic moments of neighbouring atoms tend to align due to a force called exchange coupling
d. Due to exchange coupling, atoms form domains inside which magnetic moments are aligned in the same direction

f. Graph
 χ_m
T
e. Important
 $\chi \times E$
 $1 < \mu_r < 1 + E$ (E → Small positive number)
Curie's law
$$\chi = \frac{C}{T}$$

$$M = C \frac{1}{B}$$

proportional to the absolute temperature
Magnetisation of a paramagnetic material is inversely proportional to the absolute temperature
d. Curie's law
c. Individual atom possesses permanent dipole moment
b. Eg : Al, Mn, Pt, Na, CuCl₂, O₂, Crown glass
a. Weakly attracted by a magnet
2. Paramagnetic substances

1. Diamagnetic
a. Weakly repelled by a magnet
b. Eg: Cu, Ag, Au, NaCl, H₂O etc.
c. Superconductors - Perfect conductivity
perfect diamagnetism
 $\chi = -1, \mu_r = 0$
d. Perfect diamagnetism in superconductors is called as MEISSNER EFFECT
e. Important
 $-1 < \chi < 0$
 $0 < \mu_r < 1$
f. Individual atoms do not possess permanent magnetic dipole moment
g. No effect of temperature on magnetisation

MAGNETIC MATERIALS

5) Relation between B, M and H
 $B = \mu_0 H$
 $M = \chi_m H$
also $\mu_r = 1 + \chi_m$
6) Relation between relative permeability and susceptibility
 $\chi_m = \frac{M}{H}$ also $\chi = \frac{B_m}{B_{ext}}$
scalar quantity
no unit
no dimension
SI unit → $\frac{m}{A}$ [W] → [L²A]
vector quantity
volume
also, $M = \frac{B_m}{\mu_0}$
3) Magnetisation (M)
SI unit → $\frac{A}{m}$ CGS unit → Oersted
4) Intensity of magnetizing field (H)
of medium
Relative Permeability
 $\mu_r = \frac{\mu}{\mu_0}$
Absolute Permeability
 $\mu_0 = 4\pi \times 10^{-7}$ Tesla metre / Ampere
1) Magnetic Permeability

MAGNETIC PROPERTIES

Relation between mutual inductance & self inductance

$$M = k \sqrt{L_1 L_2} \quad 0 \leq k \leq 1$$

$k \rightarrow$ coefficient of coupling
If $k=1$, Perfect flux linkage

$$M = \sqrt{L_1 L_2}$$

Otherwise $\rightarrow$ imperfect linkage
If $k=0$
no linkage
 $\Rightarrow M=0$

DYNAMIC MOTIONAL EMF DUE TO TRANSLATORY MOTION

- Charges accumulate at the ends of the conductor due to its movement in external magnetic field
- This separation of charges at the ends of the conductor causes a voltage difference

At steady state

$$F_e = F_m \quad \frac{V_{ind}}{l} = vB$$

$$e v B = e E$$

$$E = vB \quad V_{ind} \text{ or } \mathcal{E} = Blv$$

Direction to find the positive terminal
By using right hand rule,
Thumb - velocity
Fingers - Magnetic field
Palm - Positive terminal of the rod

Modification

i) Velocity is not perpendicular

$$\mathcal{E} = Blv \sin \theta$$

ii) Conductor of arbitrary shape

$\vec{v}$ Perpendicular to effective length

$$\mathcal{E} = Bl_{eff} v$$

$\vec{v}$ Parallel to effective length
 $\mathcal{E} = 0$

Semi circular loop in a magnetic field

$$I_{eff} = 2R$$

$$\mathcal{E} = Bl_{eff} v$$

$$\mathcal{E} = B \times 2Rv$$

change of area in magnetic field region

$$\Phi = BA$$

$$\Delta \Phi = B \Delta A$$

$$\mathcal{E}_{ind} = \frac{\Delta \Phi}{\Delta t} = B \frac{\Delta A}{\Delta t}$$

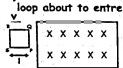
$A \rightarrow A_1 \rightarrow$ Area increases, current will be anticlockwise
 $A \rightarrow A_2 \rightarrow$ Area decreases, current will be clockwise

Shrinking Loop
Loop shrinking at rate $\frac{dr}{dt}$

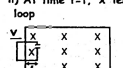
$$\mathcal{E}_{ind} = B \times 2\pi r \times \frac{dr}{dt}$$

MOTIONAL EMF : FARADAY'S LAW

i) At $t=0$ - loop about to enter



ii) At time $t=t$, 'x' length inside loop



$$A = lx$$

$$B_{ind} = B$$

$$\Phi = BA = Blx$$

$$x = vt$$

$$\Phi = Blvt$$

$$\mathcal{E}_{ind} = Blv$$

$$I_{ind} = \frac{\mathcal{E}_{ind}}{R} = \frac{Blv}{R}$$

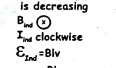
iii) Loop has fully entered



$$\mathcal{E}_{ind} = 0$$

$$I_{ind} = 0$$

iv) Loop exiting. Field is decreasing



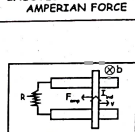
$$B_{ind} = B$$

$$I_{ind} = \text{clockwise}$$

$$\mathcal{E}_{ind} = Blv$$

$$I_{ind} = \frac{\mathcal{E}_{ind}}{R} = \frac{Blv}{R}$$

INDUCED CURRENT AND AMPERIAN FORCE



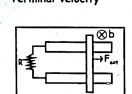
• Amperian force
 $F_{amp} = \frac{B^2 l^2 v}{R} \rightarrow$ Opposes motion

• Work done by amperian force
 $W = \Delta KE = 0 - \frac{1}{2} mv^2$ [Initial velocity, Final velocity]

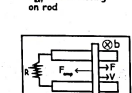
• Power developed in circuit

$$P = I^2 R \quad P = \frac{B^2 l^2 v^2}{R}$$

Terminal velocity



$t=0$ slider at rest
 F_{amp} starts acting on rod



$$t=t \quad F_{amp} = \frac{B^2 l^2 v}{R}$$

when is terminal velocity achieved $a=0$

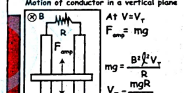
$$V = v_f \text{ constant}$$

$$F = \frac{B^2 l^2 v_f}{R}$$

$$F = mg$$

$$At \ t \rightarrow \infty \quad V_f = \frac{FR}{B^2 l^2}$$

Motion of conductor in a vertical plane



At $V = V_f$

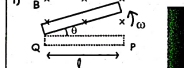
$$F_{amp} = mg$$

$$mg = \frac{B^2 l^2 v_f}{R}$$

$$V_f = \frac{FR}{B^2 l^2}$$

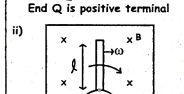
MOTIONAL EMF DUE TO ROTATIONAL MOTION

i) $B \times$



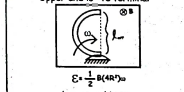
$$\mathcal{E} = \frac{1}{2} B l^2 \omega$$

End Q is positive terminal



$$\mathcal{E} = \frac{1}{2} B l^2 \omega$$

Upper end is +ve terminal



$$\mathcal{E} = \frac{1}{2} B l^2 \omega$$

Lower end is +ve

INSTANTANEOUS VALUE OF INDUCED EMF

$$\Phi = NBA \cos \omega t$$

$$= NBA \cos \omega t$$

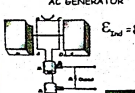
$$\mathcal{E}_{ind} = NBA \omega \sin \omega t$$

$$\mathcal{E}_0 = NBA \omega$$

when $t=0$

$$\mathcal{E}_{ind} = \mathcal{E}_0 \sin \omega t$$

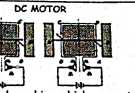
AC GENERATOR



$$\mathcal{E}_{ind} = \mathcal{E}_0 \sin \omega t$$

electrical machine which converts electrical energy into mechanical energy

DC MOTOR



electrical machine which converts electrical energy into mechanical energy

EQUATION FOR BACK EMF

$$\mathcal{E} = NBA \omega \sin \omega t$$

$$I = \frac{\mathcal{E} - NBA \omega \sin \omega t}{R}$$

MECHANICAL POWER & EFFICIENCY OF DC MOTOR

$$\eta = \frac{\text{back emf}}{\text{supply voltage}} \times \frac{P_{out}}{P_{in}} = \frac{\mathcal{E}}{V}$$

L-R CIRCUIT

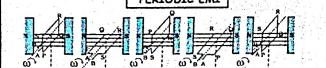
Steady state $\rightarrow$ inductor $\rightarrow$ zero resistance

For current growth in a circuit

- at $t=0$ Inductor offers infinite resistance

- at $t=\infty$ Inductor offers zero resistance

PERIODIC EMI



$$\Phi_1 = NBA \cos 0^\circ \quad \Phi_2 = NBA \cos 90^\circ \quad \Phi_3 = NBA \cos 180^\circ \quad \Phi_4 = NBA \cos 270^\circ \quad \Phi_5 = NBA \cos 360^\circ$$

$$+NBA \quad +NBA \quad -NBA \quad -NBA \quad +NBA$$

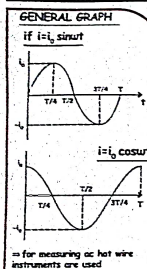
Average induced emf

$$\text{when coil is rotated from } 0^\circ \text{ to } 90^\circ \quad \mathcal{E}_{ind} = \frac{2NBA\omega}{\pi}$$

$$\text{when coil is rotated from } 0^\circ \text{ to } 180^\circ \quad \mathcal{E}_{ind} = \frac{2NBA\omega}{\pi}$$

ALTERNATING CURRENT
 "If the direction of current in a resistor or any other element changes alternately, the current is called an alternating current"

AVERAGE AND RMS VALUE OF AC
 If the current or voltage is sinusoidal then it can be expressed as
 $i = i_m \sin(\omega t + \phi)$
 $v = v_m \sin(\omega t + \phi)$
 i_m → Peak current or current amplitude
 v_m → Peak voltage or voltage amplitude
 $\omega = \frac{2\pi}{T} = 2\pi f$ T: Time period
 f : frequency (Hz or cycle/sec)
 $(\omega t + \phi)$: Total phase



ROOT MEAN SQUARE CURRENT
 $I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2 dt} = \frac{i_m}{\sqrt{2}}$
 $V_{rms} = \frac{v_m}{\sqrt{2}}$

Average value of ac is defined for positive or negative half cycle
 $\bar{I} = \frac{2I_m}{\pi}$ $\bar{V} = \frac{2V_m}{\pi}$

SAWTOOTH FUNCTION
 For half cycle $\bar{I} = \frac{I_m}{2}$
 For full cycle $\bar{I} = 0$
 Mean square current $\bar{I}^2 = \frac{I_m^2}{3}$
 rms current $I_{rms} = \frac{I_m}{\sqrt{3}}$

RECTANGULAR FUNCTION
 For half cycle $\bar{I} = I_0$
 For full cycle $\bar{I} = 0$
 Mean square current $\bar{I}^2 = I_0^2$
 rms current $I_{rms} = I_0$

AVERAGE HEAT PRODUCED DURING A CYCLE OF AC
 $H_{avg} = \frac{1}{2} I_0^2 R = I_{rms}^2 R$
 Keep in mind
 ⇒ rms value is also called virtual value or effective value
 ⇒ AC ammeter and voltmeter always measure rms value
 ⇒ Values printed on ac circuits are rms values
 ⇒ In houses ac is supplied at 220V which is the rms of voltage.
 ⇒ Peak value is $220\sqrt{2} = 311V$
 ⇒ Frequency in general is 50Hz
 ⇒ $\omega = 2\pi f = 100\pi$ rad/sec (314 rad/sec)

SINGLE COMPONENT CIRCUITS

Resistor only

- $V = V_m \sin \omega t$
- $i = i_m \sin \omega t$
- V & i are in phase
-
- $\phi = 0, \cos \phi = 1$
 $P = I_{rms} V_{rms}$
- $i_0 = \frac{V_0}{R}$ & $i_{rms} = \frac{V_{rms}}{R}$

Inductor only

- $V = V_m \sin \omega t$
- $i = i_m \sin(\omega t - \pi/2)$
- or current leads to the voltage by $\pi/2$
-
- $\phi = \pi/2, \cos \phi = 0$
- $P = 0$ (wattless circuits)
- Inductive reactance (X_L)
 $X_L = \omega L$
 Unit-ohm(Ω)
 plays role of resistance
- $i_0 = \frac{V_0}{X_L}$ & $i_{rms} = \frac{V_{rms}}{X_L}$

Capacitor only

- $V = V_m \sin \omega t$
- $i = i_m \sin(\omega t + \pi/2)$
- Current leads the voltage by $\pi/2$
-
- $\phi = \pi/2, \cos \phi = 0$
- $P = 0$ (wattless circuits)
- Inductive reactance
 $X_C = \frac{1}{\omega C}$
 Unit-ohm(Ω)
 plays role of resistance
- $i_0 = \frac{V_0}{X_C}$ & $i_{rms} = \frac{V_{rms}}{X_C}$

SUMMARY

	Z (Impedance)	ϕ
1. R only	R	0
2. L only	$X_L = \omega L$	$-\pi/2$
3. C only	$X_C = \frac{1}{\omega C}$	$\pi/2$

AVERAGE VALUE OF AC FOR ONE TIME PERIOD
 $\bar{I} = \frac{1}{T} \int_0^T i dt = \frac{1}{T} \int_0^T i_m \sin \omega t dt$ → area of $i-t$ graph / time
 $\bar{I} = 0$ for $0 \rightarrow T$ for a sinusoidal wave.
 The average value of sin or cos function for one time period or n time periods ($n=1, 2, \dots$) is zero
 Keep in mind
 Long period is equivalent to one time period

PHASOR DIAGRAM
 Diagram representing ac voltage or current as vectors with phase angle between them.
 $i = i_m \sin(\omega t)$
 $v = v_m \sin(\omega t + \phi)$

Mean square current for one time period
 $\bar{i}^2 = \frac{1}{T} \int_0^T i^2 dt = \frac{i_m^2}{2}$
 Remember
 The average value of square of sin or cosine function for one time period is $\frac{1}{2}$
 $\int_0^{2\pi} \sin^2 \omega t d\omega t = \frac{1}{2} \times 2\pi$
 $\int_0^{2\pi} \cos^2 \omega t d\omega t = \frac{1}{2} \times 2\pi$

AVERAGE POWER CONSUMPTION

$P = VI \cos \phi$
 for $\cos \phi = 1$ or $\phi = 0$
 $P_{avg} = V_{rms} I_{rms}$

1) R-L CIRCUITS

- $V = V_m \sin \omega t$ $V_R = i_m R$ $V_L = i_m X_L$
- Voltage phasor diagram
 $V = \sqrt{V_R^2 + V_L^2}$
 $\tan \phi = \frac{V_L}{V_R}$
- $i = i_m \sin(\omega t - \phi)$
- $\frac{V_0}{i_0} = \frac{V_{rms}}{i_{rms}} = Z = \sqrt{R^2 + X_L^2}$

SERIES AC CIRCUITS

5. Impedance phasor
 $Z = \sqrt{R^2 + X^2}$
 $\tan \phi = \frac{X}{R}$

6. $i_0 = \frac{V_0}{Z}$

2) R-C CIRCUITS

- $V = V_m \sin \omega t$ $V_R = i_m R$ $V_C = i_m X_C$

2. Voltage phasor diagram
 $V = \sqrt{V_R^2 + V_C^2}$
 $\tan \phi = \frac{V_C}{V_R}$

3. $i = i_m \sin(\omega t + \phi)$

4. Impedance phasor
 $Z = \sqrt{R^2 + X_C^2}$
 $\tan \phi = \frac{X_C}{R}$

5. $i_0 = \frac{V_0}{Z}$
 or $i_{rms} = \frac{V_{rms}}{Z}$